TECHNICAL PRESENTATION

CSA0509 – Database Management Systems
CO1 – Assessment Tool 3

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ER Diagram

Definition

- Graphical representation of a database.
- Shows Entities, Attributes, and Relationships.
- Used before database implementation.

Objectives

- Organise data
- Reduce redundancy
- Improve consistency
- Simplify database design

ER Diagram Components

Entity

- Real-world object
- Example: Member, Book

Attribute

- Describes an entity
- Example: Name, BookID

Relationship

- Connects entities
- Example: Borrows, Issues

Keys

- Primary Key (PK)
- Foreign Key (FK)

ER Diagram for an Online Library System

Entities

- Member
- Book
- Librarian

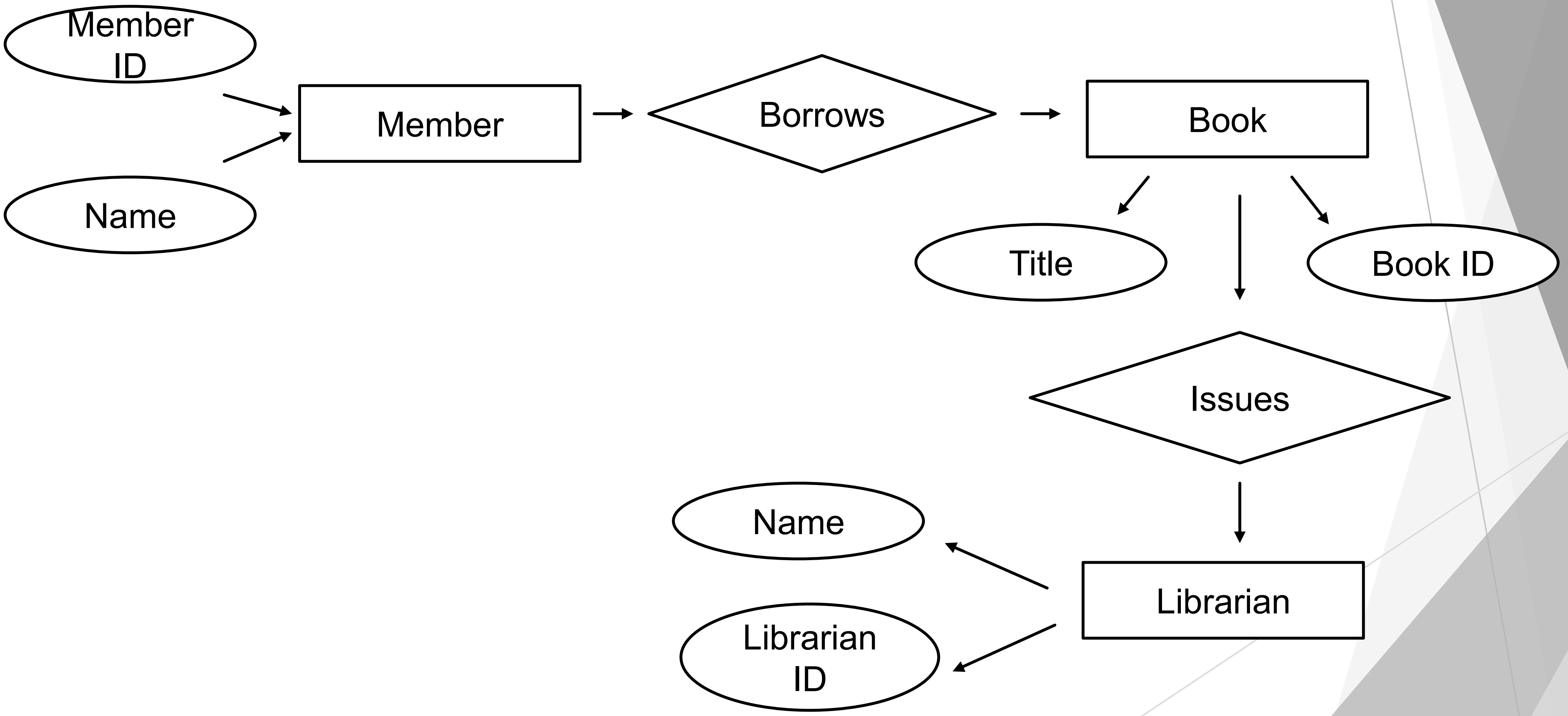
Relationships

- Member → Borrows → Book
- Librarian → Issues → Book

Advantages

- Easy database design
- Maintains integrity
- Reduces duplication
- Efficient data retrieval

ER Diagram for an Online Library System



Enhanced Entity Relationship (EER) Model

Definition

- Extension of the ER Model.
- Represents complex real-world data.
- Supports inheritance.

Advantages

- Better data modelling
- Reduces redundancy
- Improves database flexibility
- Supports inheritance

Generalization

Definition

- Combines similar entities into one
- Common attributes are stored in the superclass.

Benefits

- Eliminates duplicate data
- Improves consistency
- Easier maintenance.

University Example

Student

Faculty

Staff

↓

Person (Superclass)

Specialization

Definition

- Divides one Superclass into multiple Subclasses.
- Each subclass has unique attributes.

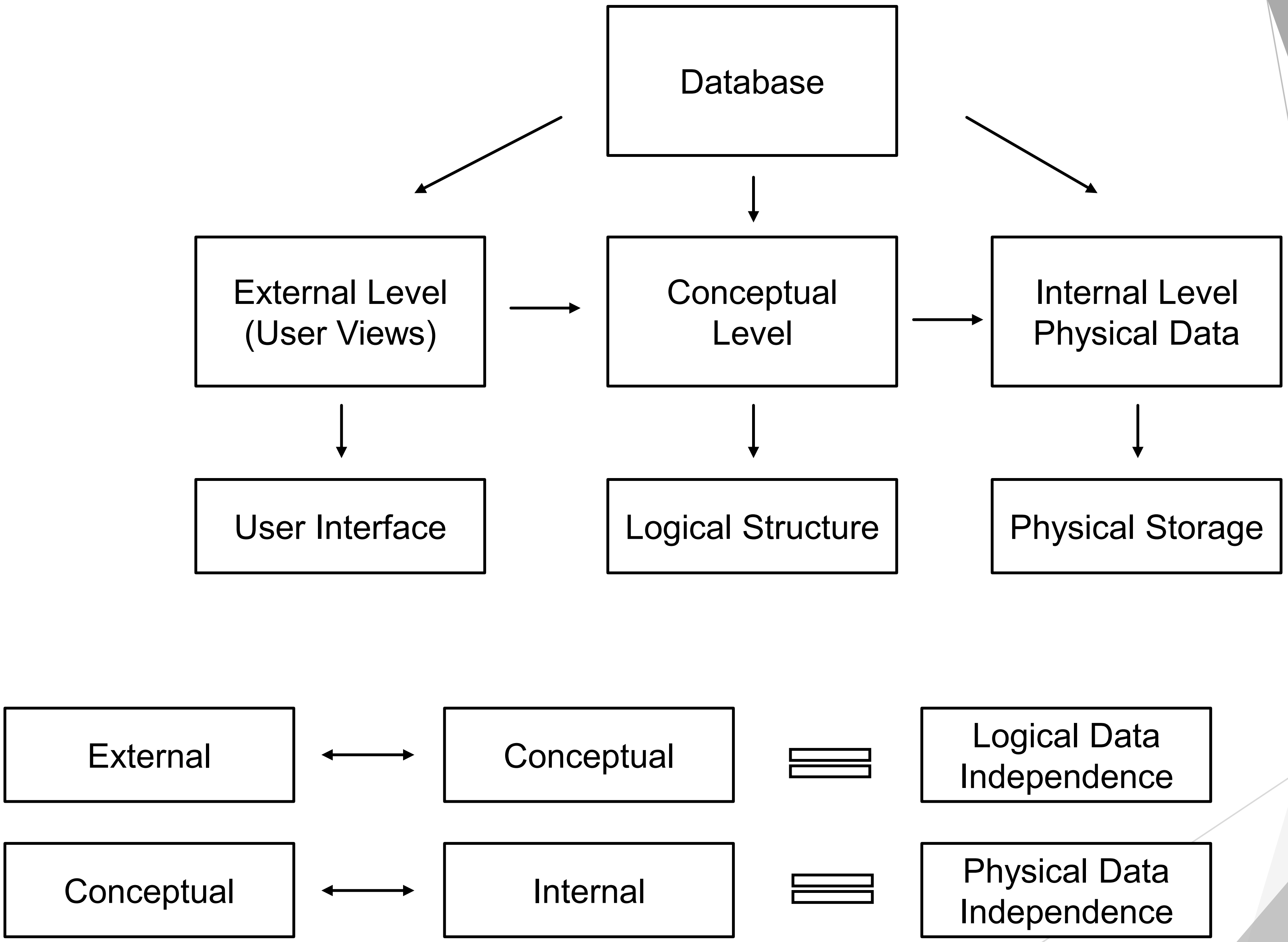
Applications

- Student Management
- Faculty Records
- HR Management

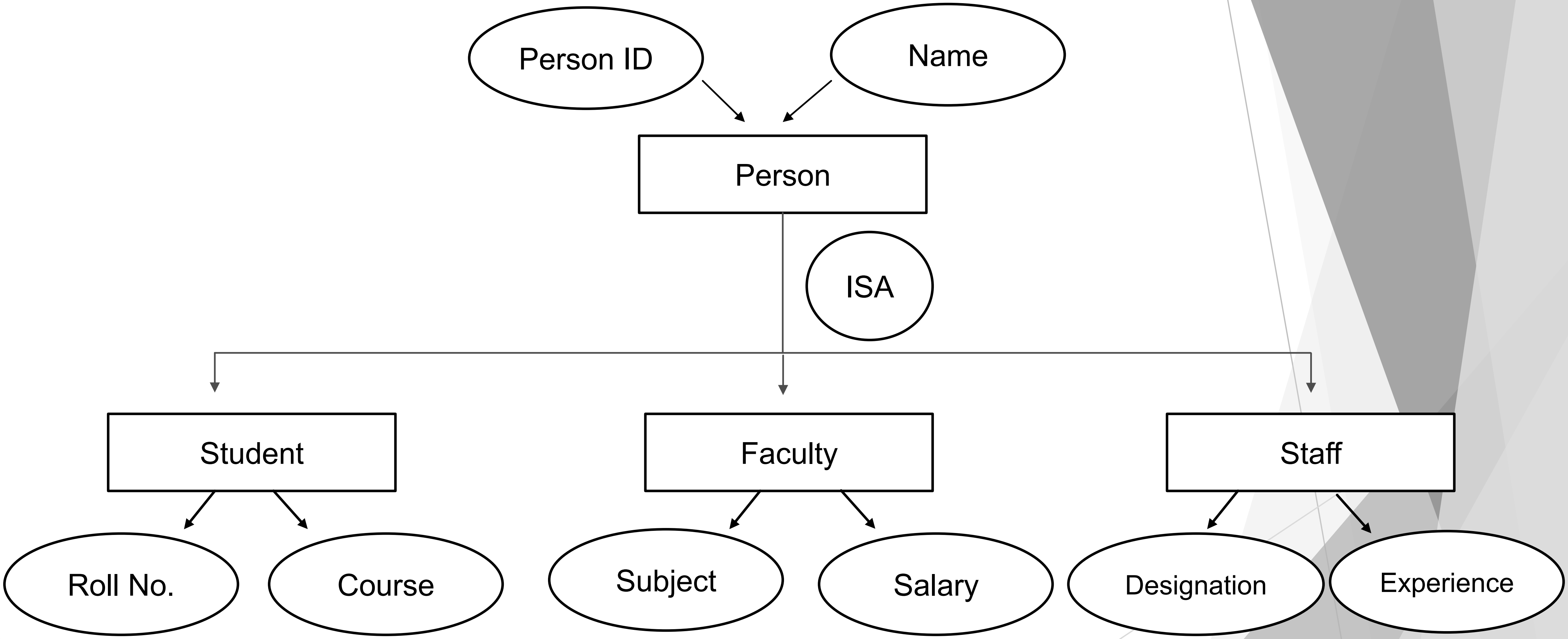
University Case Study
Person

- Student
- Faculty
- Staff

Student → Roll No., Course
Faculty → Salary, Subject
Staff → Designation


Database
External Level
(User Views)
Conceptual
Level
Internal Level
Physical Data
User Interface
Logical Structure
Physical Storage
External
Conceptual
Logical Data
Independence
Conceptual
Internal
Physical Data
Independence

EER Diagram (University System)



Data Independence

Definition

- Ability to modify one level of the database without affecting another level.
- Ensures flexibility in database management.

Types

Logical Data Independence
Physical Data Independence

Importance

- Easy maintenance
- Better flexibility
- Supports database updates
- Reduces application changes

Types of Data Independence

Logical Data Independence

- Changes the Conceptual Schema.
- Does not affect user applications.

Example

Adding a new column to a table.

Physical Data Independence

- Changes the Internal Storage.
- Does not affect the logical structure.

Example

Changing file organisation or indexing.

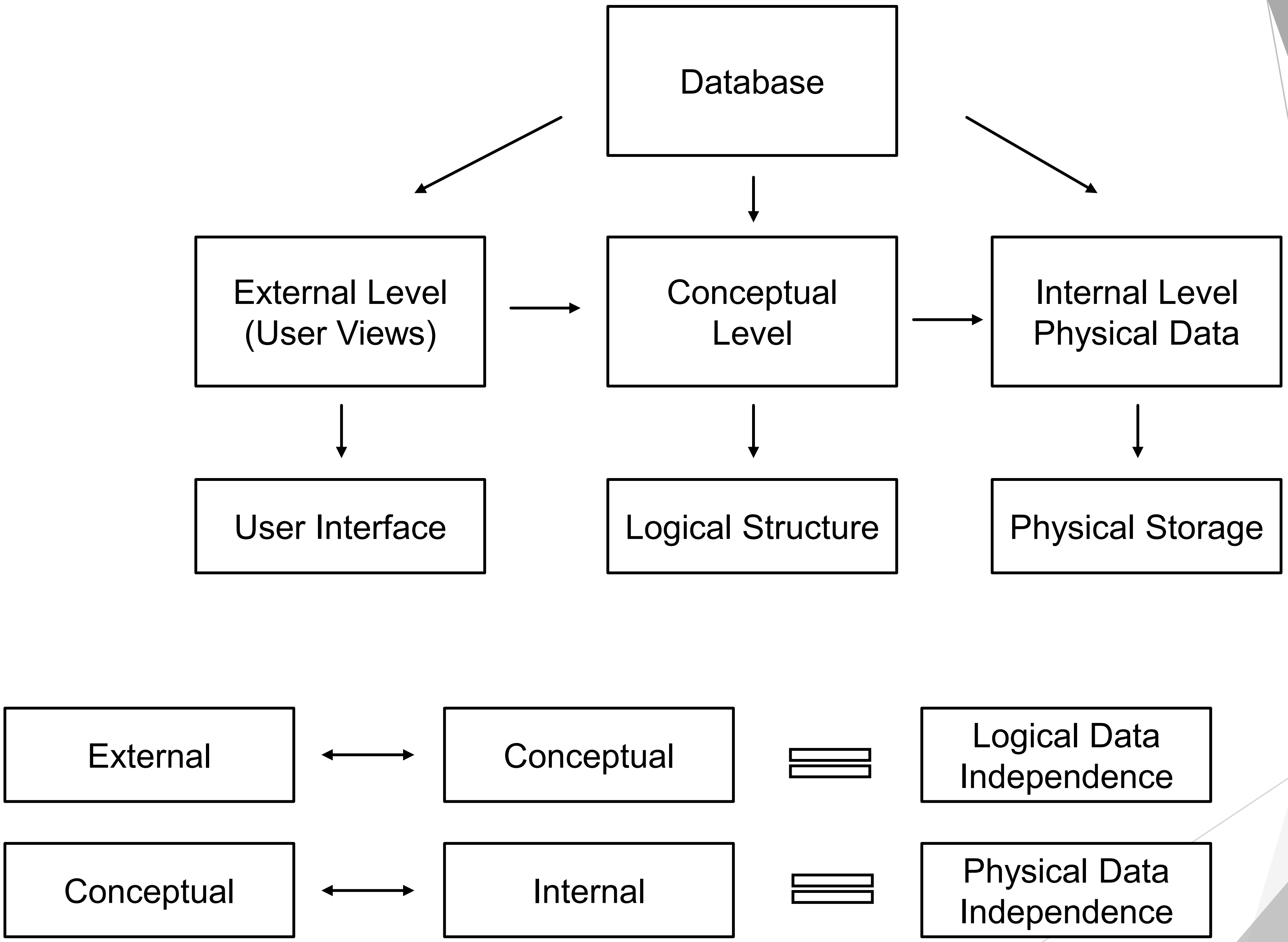
Significance in DBMS Architecture

Benefits

- Easy database modification
- Improved performance
- Better data security
- Reduced maintenance cost
- Supports scalability

Real-Life Applications

- Banking System
- Hospital Management
- Library Management
- University Database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Software Components of DBMS

Definition

- DBMS consists of software components that manage database operations.
- These components process queries, store data, and maintain transactions.

Main Components

- Query Processor
- Storage Manager
- Transaction Manager

Functions

- Query execution
- Data storage
- Transaction management
- Database security

Query Processor & Storage Manager

Query Processor

- Interprets SQL queries.
- Optimises query execution.
- Retrieves requested data.

Storage Manager

- Manages database files.
- Controls memory buffers.
- Organises physical storage.

Advantages

- Faster query execution
- Efficient data access
- Better storage management

Transaction Manager

Definition

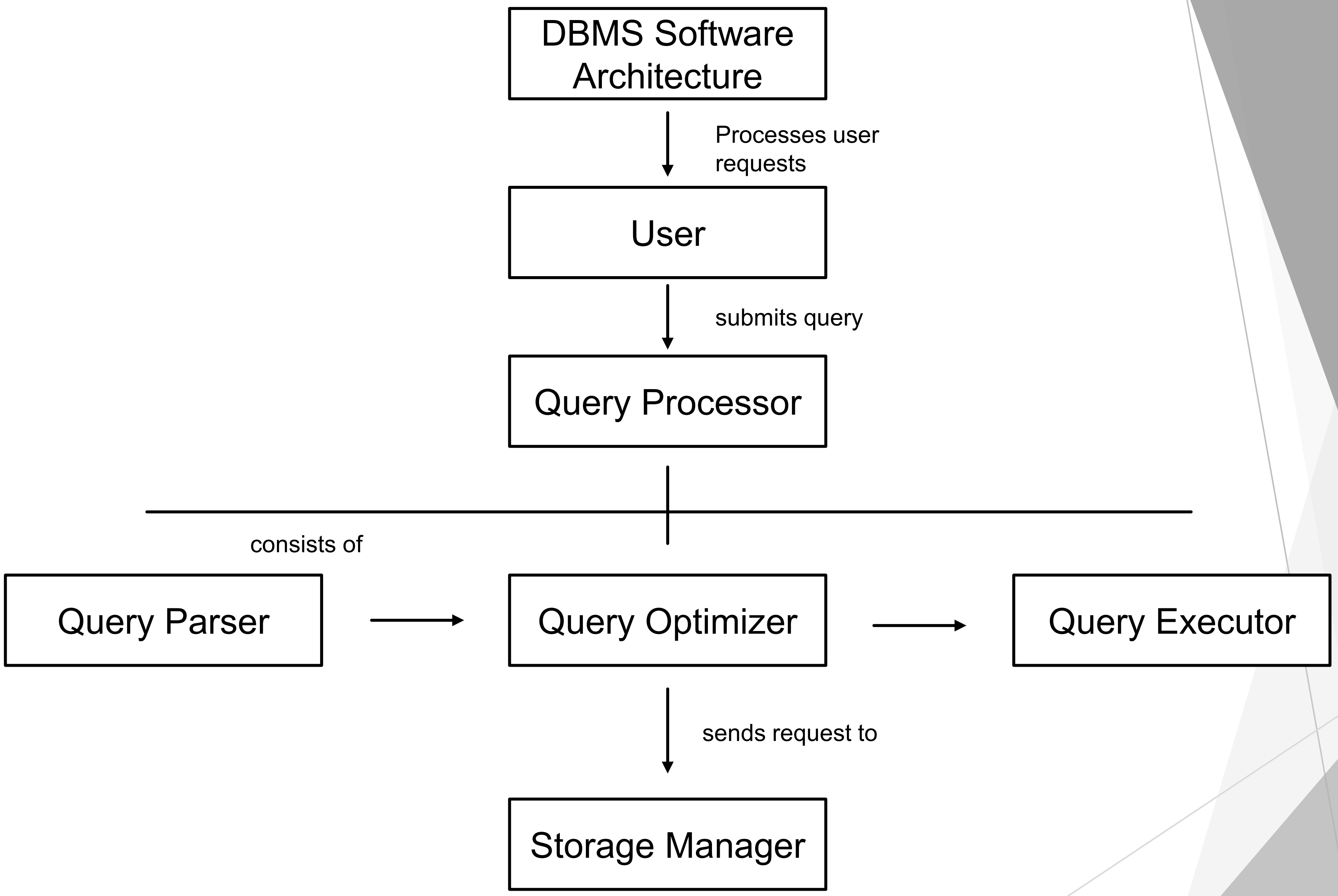
- Controls all database transactions.
- Ensures data consistency.
- Maintains ACID properties.

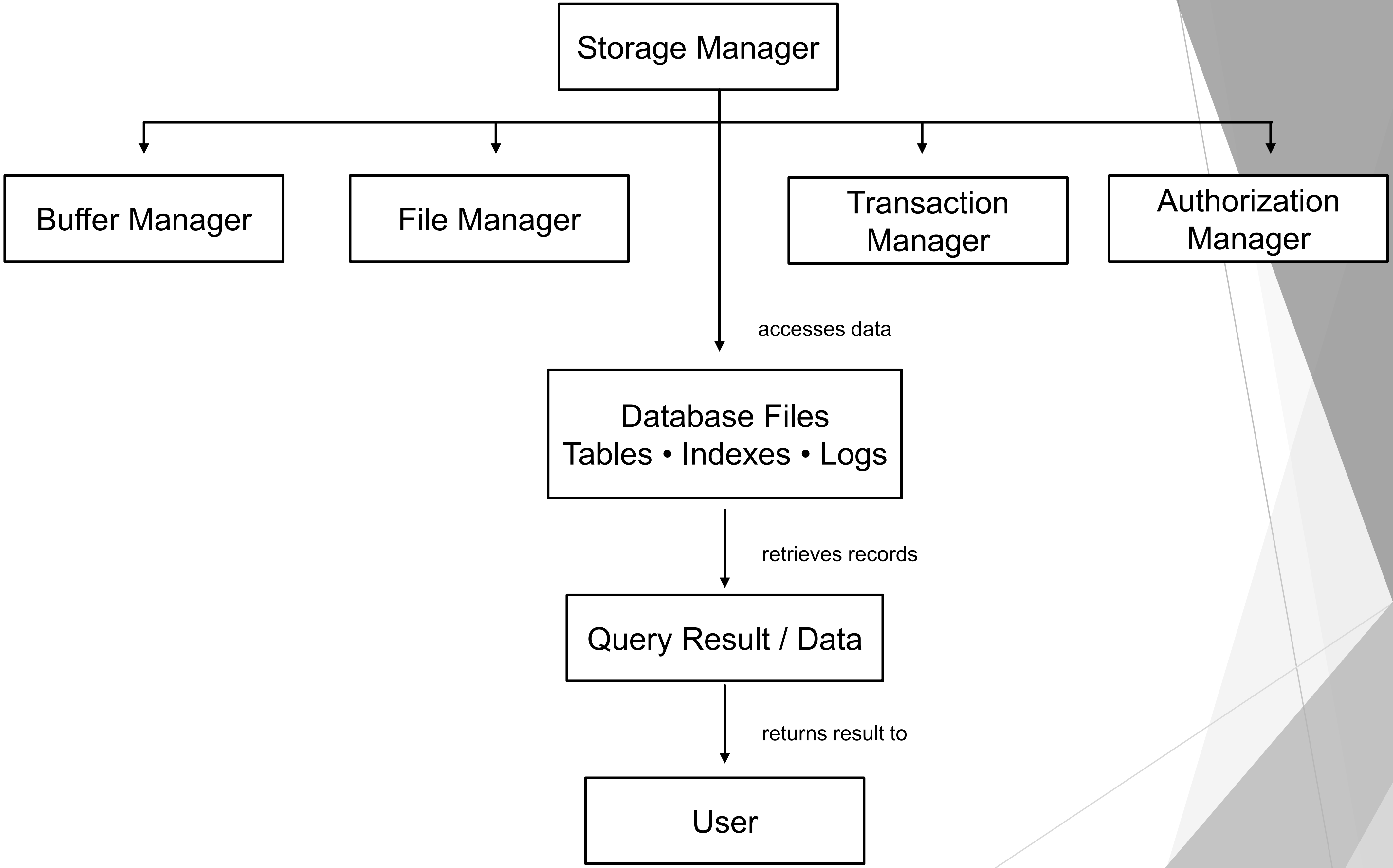
Responsibilities

- Concurrency Control
- Recovery Management
- Rollback & Commit
- Error Handling

Applications

- Banking System
- ATM Transactions
- Online Shopping
- Railway Reservation


DBMS Software Architecture
Processes user requests
User
submits query
Query Processor
consists of
Query Parser
Query Optimizer
Query Executor
sends request to
Storage Manager


Storage Manager
Buffer Manager
File Manager
Transaction Manager
Authorization Manager
accesses data
Database Files
Tables • Indexes • Logs
retrieves records
Query Result / Data
returns result to
User

Strong and Weak Entities

Definition

- An Entity represents a real-world object in a database.
- Entities are classified as:
    - Strong Entity
    - Weak Entity

Importance

- Identifies database objects.
- Maintains relationships.
- Improves database design.
- Ensures data integrity.

Strong Entity

Definition

- Has its own Primary Key.
- Exists independently.

Characteristics

- Independent entity
- Unique identification
- Single rectangle representation

Example
Student

Student_ID (PK)
Name
Department

Weak Entity

Definition

- Does not have a Primary Key.
- Depends on a Strong Entity.

Characteristics

- Dependent entity
- Uses a foreign key
- Double rectangle representation

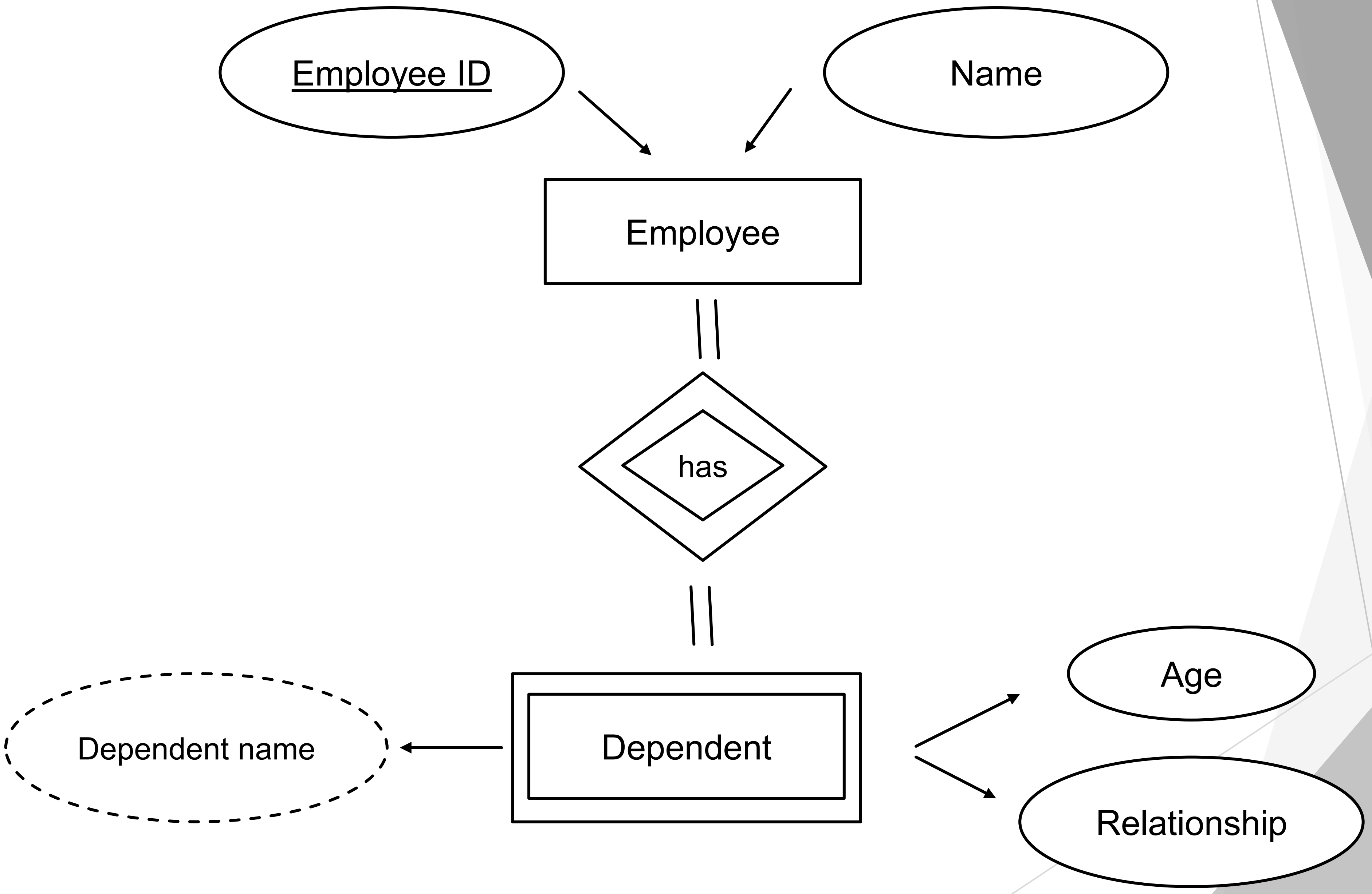
Example

Dependent

Dependent_Name

Age

Depends on Employee



Employee ID
Name
Employee
has
Dependent name
Dependent
Age
Relationship

THANK YOU